

Environmental Economics (ECON 4150)

Fall 2026 syllabus

This course is about the economic foundations of global environmental problems, including air and water pollution, with discussions of alternative (command and market based) solutions.

When	Tuesday and Thursday 4:35-5:55pm
Where	Benson Hall C115
Who	Dr. Emmett Reynier, Assistant Professor
Email	emmettr@uga.edu
Office	Amos Hall B324
Office hours	Tuesday 10-11am, Wednesday 2-3pm
Prerequisite	Intermediate Microeconomics, ECON 4010

Course overview

What is the economic value of a clean river or a reduction in greenhouse gas emissions? How do we design policies to tackle climate change while minimizing economic costs? What role do markets play in environmental degradation, and how can they be leveraged to improve environmental outcomes?

In this course, we will explore the intersection of economics and environmental policy. You'll learn how economic theory applies to environmental issues, examining both the causes of environmental problems and the potential solutions. We'll cover topics like externalities, public goods, and the valuation of environmental goods and services. Additionally, we'll analyze various policy instruments, including taxes, subsidies, and market-based mechanisms like cap-and-trade, that economists use to address environmental challenges. By the end of the course, students will have developed a solid foundation in environmental economics, gaining insights into how economic tools can be applied to create effective, equitable, and efficient environmental policies. Expect to leave with a better understanding of both theoretical concepts and their practical applications in the real world.

Course structure

Successful completion of the course will entail a mix of in-class participation, bi-weekly problem sets, midterm and final exams, and a final project. More details on each of these are below.

Grades

Component	Weight
Final project	30%
Final exam	30%
Midterm	20%
Participation	10%
Problem sets	10%

A class participation bonus worth an additional 2.5% may be awarded; see the description below.

We will return grades on all submitted material within one week from its due date. Please note that you are going to be graded on a curve. This means that the absolute scores or percentages from your midterm and final are largely irrelevant. What matters most is where you are in the distribution of scores among your peers. Your letter grades that I post at the end of the semester will reflect this curve. These are the *only* criteria by which you will be graded. I will not consider any additional submissions/essays/tests/etc. to change your final grade.

Lectures

I will post lecture slides on eLC ahead of time. These slides are designed to be self-contained, meaning they will cover everything you need to succeed in the course. However, they will not be fully complete; I reserve some of the key material for the in-class lecture, where we will work through it together. This is to encourage active participation, note-taking, and deeper understanding. You'll need to be present in class to capture this content and benefit from our discussions.

I expect you to be engaged during each lecture. This means asking questions when something is unclear and contributing to class discussions. Engaged students help create a more dynamic and enjoyable learning environment, so don't hesitate to share your thoughts or ask for clarification. Participation will be rewarded in two ways:

- 10% of your grade is based on class participation. For every class you attend and are actively engaged in, you'll earn 0.5 percentage point (up to a maximum of 10 points).
- You can also earn a 2.5 percentage point bonus by bringing in a current news article that connects with a topic we're discussing. This is a great way to link classroom concepts to real-world events, and I encourage you to take advantage of it.

Attendance policy

In-person attendance is important for your learning in this class! However, **you do not need to tell me if you are missing a particular lecture** — by only requiring you to participate in 20 classes, I am equitably distributing the ability to miss class in case of an emergency, religious holiday, illness, etc. Please do not come to class if you are sick.

Problem sets

There will be six problem sets throughout the semester. These problem sets are designed to give you the opportunity to practice the types of problems that you will see on the midterm and final exam, and to reinforce concepts from the lectures; particularly the sections and examples that we work through together during class.

All problem sets will be posted as assignments on eLC. You are welcome to hand-write or type your answers, but will be required to submit your assignments on eLC. We will drop your lowest problem set score. We will accept late problem sets up to 48 hours after the assignment due date, but will subtract 1 percentage point from your grade for each hour that it is late. No assignments will be accepted after the solution key has been posted.

Examinations

The midterm exam is **Thursday, Oct 8th 4:35 pm – 5:55 pm**. Please note that no make-up midterm will be given. If you know that you cannot take the midterm for valid, non-medical reasons (e.g. sports/athletic events), talk to me immediately. Failure to do so will result in a zero grade. If you miss the midterm due to a medical or last-minute emergency, then the weighting for this section of the course will be placed on the final exam.

The final exam is **Thursday, Dec 3rd 3:30 pm – 6:30 pm**. It will be a cumulative exam, i.e. covering the entire syllabus. Make-up final exams will be allowed in the case of an emergency with advanced notice. Missing the final exam without advanced notice will result in a zero grade. If you know now that you cannot make this exam, do not register for the class.

Both the midterm and final exam will be closed notes, closed book. Acceptable items are limited to: pens/pencils and a straight-edge for the final exam, and non-programmable, non-cell phone based hand-held calculators.

Final project

You will complete a project in groups of 3 to 4. The main deliverable for will be a paper (7–10 pages) and an accompanying presentation (8–12 minutes) to the class on either **November 17th and 19th**.

I will post a full description to eLC. You will pick a topic that you are interested in, do a brief review of the existing economics literature on that topic, collect some relevant data, and do preliminary analysis. There will be several project milestones throughout the term to ensure that you are making consistent progress.

Course materials

There is no required textbook for this course. The lecture slides are the primary material, and they are designed to be self-contained. Any readings will be linked directly or posted to eLC.

For motivated students who want to deepen their understanding of environmental economics, here are two optional books you might consider:

- “**Markets and the Environment**” (2nd edition, ISBN 9781610916073) by Keohane and Olmstead (K&O). I’m a big fan of this book. It does an excellent job of conveying the intuition of environmental economics and discusses a bunch of interesting case studies. It’s also pretty cheap and has proven popular with students in the past.
- “**Environmental and Natural Resource Economics**” (12th edition, ISBN 9781032101187) by Tietenberg and Lewis (T&L). Another well-regarded textbook that provides a solid, if superficial, introduction to the field.

Course policies and expectations

No laptops, tablets, or cell phones during the lecture. A **2024 meta-analysis of 27 randomized experiments** finds a large negative effect of phone distraction on how much of a lecture students retain, and a **randomized trial across 50 sections of introductory economics** found that merely *permitting* laptops and tablets lowered final exam scores by 0.18 standard deviations. The cost is not only yours: students seated in view of someone else’s screen **do worse too**. I may make exceptions in relevant cases, just ask.

I'd like to encourage an interactive and engaging classroom atmosphere. In-class participation will be rewarded, and you should expect me to call on you to answer questions and discuss ideas during lectures. I will treat you as adults and anticipate that you can engage with challenging or uncomfortable ideas accordingly. At the same time, discriminatory or egregiously inflammatory language will not be tolerated. Similarly, the university takes an appropriately hard-line policy on sexual discrimination and violence with which you should acquaint yourself.

Communication

Any information that I need to tell the whole class (e.g. posting a problem set), I will send via an announcement in eLC. Make sure you check your notification settings to ensure that you see these in a timely manner. Individual communication with students will be through email.

If you have...

- **A practical yes/no type question** about an assignment, lecture, or other component of our class, please ask it via the "Class Question and Answers" discussion on eLC. Students often have the same questions, and so this can be a resource for you to get answers quicker. I will monitor the discussion regularly, but answer yourself if you see an unanswered question you know the answer to! Remember to be respectful of your peers in any responses.
- **A technical question about eLC**, please use the [UGAs eLC Help Page](#).
- **A question about course content or activities, about something personal, time sensitive, or something else** that doesn't feel like it fits above, please reach out to me by email or by attending office hours! I welcome meetings outside my regular office hours as well, just email me to set a time.

I encourage you to reach out whenever you have questions! Whether it's clarifying course material, connecting what we discuss in class to current events, or exploring how the field of environmental economics can shape your career, I'm here to help. My goal is to create a welcoming environment where you feel comfortable discussing any challenges or curiosities you have. Whether you're seeking advice on course topics, curious about applying concepts to real-world situations, or looking for career guidance, my door is always open.

During office hours, students bring in a wide range of concerns, questions, and successes. We might talk through a specific concept or problem to clarify it, might think together about an issue a student is curious about that relates to the class, might discuss a student's post-graduation goals, might identify more effective ways to study for a future exam or to begin a project, or any number of other topics. Some students have never been to an instructor's office hours — if that is you, please change that this semester by attending! Feel free to come with a peer if that is useful for you.

Environmental Awareness Policy Criteria

To achieve the goals of the Environmental Awareness (EA) Requirement, every University of Georgia undergraduate student must acquire a basic understanding of:

- a) The interactions between human activity and the environment at local, regional, or global scales; OR
- b) The ethical, cultural, economic, or political forces that affect environmental problems and policies.

This course satisfies the University of Georgia's Environmental Awareness Requirement. Specifically, this course provides students with an understanding of the connections between environmental quality and the economic behavior of consumers, producers, and government at local, regional, and global scales. This course also provides students with an understanding of economic forces that affect environmental problems and policies that attempt to correct market failures associated with environmental quality.

Use of AI

You are allowed to use AI tools for this class under certain restrictions. I find AI tools useful in my own research and want to help you learn how to best utilize them for your own learning. However the key word in that sentence is **learning**, which requires time and effort. I strongly caution against outsourcing all of that time and effort to an LLM. Practically, you will have to demonstrate an understanding of the material during the midterm, final, and project presentations without access to AI. More generally, the job market will not reward parroting what AI tells you. That might be a scary proposition, but at the same time there are huge opportunities to learn and do cool things without many of the constraints that used to exist.

The rules:

1. All chat conversations must begin with the prompt below. Any agentic AI tool you use for the class must have the AGENTS.md file I've uploaded to eLC.
2. Answers to problem sets and the written project cannot be written by AI. You are welcome to use AI to write and debug code for the project — writing code is not the skill I am assessing. Deciding what analysis to run, and working out what it means, is.
3. No AI on the midterm or the final. Both are closed book and closed device, so this should not come up — but to be unambiguous, using a tool during either is academic dishonesty.
4. All submissions (problem sets and the project) must contain a disclosure at the end describing how you used AI. Tell me three things: which tool, what you asked it for, and what you did with what it gave back. Two or three sentences is

- plenty. “I did not use AI on this assignment” is a complete disclosure.
5. You are responsible for everything you submit. If a tool invents a citation, has a bug in its code, or gets the economics wrong, that is your error on your assignment.

I will treat breaking these rules as a violation of the UGA Student Honor Code.

You are not required to use AI in this course, and you are never required to pay for it. You can succeed in every assignment for this class without ever opening a chatbot. If you do want to use one, UGA already provides Gemini, NotebookLM, and Microsoft Copilot to all students at no cost, see [UGA's AI tools page](#).

Here is the prompt:

I'm a student in ECON 4150, Environmental Economics at UGA. Act as a tutor, not an answer key. Explain concepts, ask me guiding questions, and critique work I've already written. You can write and debug code once I've told you what it should do, and you can summarize papers I give you — but I decide what analysis to run, what the results mean, and how the papers fit together. Do not write prose I'll submit, produce final numeric answers, carry out derivations or draw diagrams I've been assigned, or invent citations. If I ask for any of that, decline and help me do it myself instead. I'm responsible for anything you get wrong, I have to disclose how I used you, and I'll be tested on this material closed-book.

I am no AI expert, but during the term I'll be providing guidance on what I have learned about using it effectively. For transparency, I will be using AI to assist with some portions of this class: (1) Cleaning up the content and visuals in my lecture slides, and (2) Generating variants from my existing question bank for problem sets and tests. I will never use AI to grade assignments, nor to read or respond to your emails/messages.

UGA Student Honor Code

“I will be academically honest in all of my academic work and will not tolerate academic dishonesty of others.” A Culture of Honesty, the University's policy and procedures for handling cases of suspected dishonesty, can be found at integrity.uga.edu.

Accommodation for disabilities

If you plan to request accommodations, please register with Accessibility and Testing. They can be reached by visiting Clark Howell Hall, calling 706-542-8719 (voice) or 706-542-8778 (TTY), or by visiting <https://accessibility.uga.edu/>.

UGA Well-being Resources

UGA Well-being Resources promote student success by cultivating a culture that supports a more active, healthy, and engaged student community. Anyone needing assistance is encouraged to contact Student Care & Outreach (SCO) in the Division of Student Affairs at 706-542-8479 or visit sco.uga.edu. Student Care & Outreach helps students navigate difficult circumstances by connecting them with the most appropriate resources or services. They also administer the Embark@UGA program which supports students experiencing, or who have experienced, homelessness, foster care, or housing insecurity. UGA provides both clinical and non-clinical options to support student well-being and mental health, any time, any place. Whether on campus, or studying from home or abroad, UGA Well-being Resources are here to help.

- Well-being Resources: well-being.uga.edu
- Student Care and Outreach: sco.uga.edu
- University Health Center: healthcenter.uga.edu
- Counseling and Psychiatric Services: caps.uga.edu or CAPS 24/7 crisis support at 706-542-2273
- Health Promotion/ Fontaine Center: healthpromotion.uga.edu
- Accessibility & Testing: accessibility.uga.edu

Additional information, including free digital well-being resources, can be accessed through the UGA app or by visiting well-being.uga.edu.

Statement on Inclusive Excellence

The Terry College of Business is committed to promoting an inclusive learning and working environment among its students, faculty, and staff. This class welcomes the open exchange of ideas and values freedom of thought and expression and provides a professional environment that recognizes the inherent worth of every person. It aims to foster dignity, understanding, and mutual respect among all individuals in the class.

The course syllabus is a general plan for the course; deviations announced to the class by the instructor may be necessary.

Course outline

Here is a **tentative** schedule for the semester.

Wk	Dates	Topic	K&O	T&L
1	Aug 18–20	Introduction and Economic efficiency	1	1
2	Aug 25–27	Externalities and market failures	2, 4, 5	2
3	Sep 1–3	Evaluating trade-offs: cost-benefit analysis	3	3
4	Sep 8–10	Non-market valuation: stated preferences	3	4
5	Sep 15–17	Non-market valuation: revealed preferences	3	4
6	Sep 22–24	Pollution control: allocations	8, 9, 14	7
7	Sep 29 – Oct 1	Pollution control: taxes and permits	8, 9, 14	7
8	Oct 6–8	Review; midterm Oct 8		
9	Oct 13–15	Air pollution	10	8, 15
10	Oct 20–22	Water pollution	9, 10	9
11	Oct 27–29	Climate change: impacts	10	11, 12
12	Nov 3–5	Climate change: adaptation	10	13, 14
13	Nov 10–12	Trade and Environmental justice		10
14	Nov 17–19	Research paper presentations		
15	Nov 24–26	No class — Thanksgiving break		
16	Dec 1–3	Review; final exam Thu Dec 3, 3:30–6:30pm		

Chapter numbers refer to the two optional texts described under *Course materials*. Nothing in those columns is required reading.